Tabela surowych próbek dyskretnych – Trend MF-BB z dwustrefową funkcją celu

Zestawienie dyskretnych próbek trajektorii czasowych (t, v_C, i_L, s) dla regulatora Trend MF-BB nastrojonego dwustrefową funkcją celu prof. Iwańskiego ($w_i^* = 0.3583, w_v^* = 6.6502, f_{c,i}^* = 687.2\text{ Hz}, f_{c,u}^* = 250.9\text{ Hz}$, $T_s = 10\ \mu\text{s}$).

Okno 1: Stan ustalony pod pełnym obciążeniem (240 V, 1.2 kW, t □ [145.0, 146.5] ms, krok 20 µs)

t [ms]	v_C [V]	i_L [A]	s	t [ms]	v_C [V]	i_L [A]	s
145.000	239.98	11.604	1	145.760	239.67	13.155	1
145.020	239.48	14.254	1	145.780	239.85	12.608	0
145.040	240.22	10.507	0	145.800	239.93	12.059	1
145.060	239.71	13.158	1	145.820	239.42	14.708	1
145.080	239.89	12.611	0	145.840	240.21	10.962	0
145.100	240.47	8.858	0	145.860	240.21	10.411	1
145.120	239.96	11.511	1	145.880	239.70	13.062	1
145.140	239.46	14.160	1	145.900	239.87	12.515	0
145.160	240.19	10.415	0	145.920	239.95	11.966	1
145.180	239.68	13.066	1	145.940	239.45	14.615	1
145.200	239.86	12.519	0	145.960	240.22	10.868	0
145.220	239.93	11.970	1	145.980	240.22	10.317	1
145.240	239.43	14.619	1	146.000	239.71	12.968	1
145.260	240.20	10.873	0	146.020	239.88	12.421	0
145.280	240.20	10.322	1	146.040	239.95	11.872	1
145.300	239.70	12.974	1	146.060	239.45	14.521	1
145.320	239.86	12.427	0	146.080	240.22	10.775	0
145.340	239.94	11.878	1	146.100	240.21	10.224	1
145.360	239.43	14.527	1	146.120	239.70	12.876	1
145.380	240.20	10.781	0	146.140	239.86	12.329	0
145.400	239.69	13.432	1	146.160	239.93	11.780	1
145.420	239.88	12.884	0	146.180	239.43	14.430	1
145.440	240.49	9.131	0	146.200	240.19	10.684	0
145.460	239.98	11.783	1	146.220	239.68	13.335	1
145.480	239.48	14.433	1	146.240	239.87	12.788	0
145.500	240.23	10.686	0	146.260	240.46	9.035	0
145.520	240.22	10.135	1	146.280	239.95	11.688	1
145.540	239.72	12.786	1	146.300	239.45	14.337	1
145.560	239.87	12.240	0	146.320	240.20	10.591	0
145.580	239.94	11.691	1	146.340	239.69	13.242	1
145.600	239.43	14.340	1	146.360	239.87	12.695	0
145.620	240.18	10.595	0	146.380	240.46	8.942	0
145.640	239.68	13.246	1	146.400	239.95	11.595	1
145.660	239.86	12.699	0	146.420	239.45	14.244	1
145.680	240.44	8.946	0	146.440	240.19	10.499	0
145.700	239.94	11.599	1	146.460	239.68	13.150	1
145.720	239.43	14.249	1	146.480	239.86	12.603	0
145.740	240.17	10.504	0	146.500	239.94	12.054	1

Okno 2: Stan przejściowy – skok zadanego napięcia 240 V → 160 V (t □ [149.8, 151.3] ms, krok 20 µs)

t [ms]	v_C [V]	i_L [A]	s	t [ms]	v_C [V]	i_L [A]	s
149.800	239.44	14.298	1	150.560	204.32	-18.986	1
149.820	240.19	10.552	0	150.580	202.98	-19.010	0
149.840	239.68	13.203	1	150.600	200.52	-21.697	0
149.860	239.86	12.656	0	150.620	200.10	-19.003	1
149.880	239.95	12.107	1	150.640	198.77	-18.971	0
149.900	239.44	14.756	1	150.660	196.33	-21.546	0
149.920	240.23	11.009	0	150.680	195.92	-18.853	1

t [ms]	v_C [V]	i_L [A]	s	t [ms]	v_C [V]	i_L [A]	s
149.940	240.23	10.458	1	150.700	194.60	-18.764	0
149.960	239.73	13.109	1	150.720	192.20	-21.230	0
149.980	239.90	12.561	0	150.740	191.79	-18.536	1
150.000	240.47	8.808	0	150.760	189.42	-20.928	0
150.020	240.66	5.049	0	150.780	187.95	-20.750	1
150.040	240.48	1.296	0	150.800	187.56	-18.057	1
150.060	239.92	-2.444	0	150.820	185.25	-20.337	0
150.080	239.00	-6.158	0	150.840	183.82	-20.105	1
150.100	237.70	-9.838	0	150.860	183.43	-17.413	1
150.120	236.04	-13.474	0	150.880	181.20	-19.585	0
150.140	234.02	-17.056	0	150.900	178.76	-21.692	0
150.160	231.66	-20.574	0	150.920	178.39	-18.998	1
150.180	230.10	-20.960	1	150.940	177.11	-18.676	0
150.200	229.62	-18.267	1	150.960	174.77	-20.676	0
150.220	227.15	-21.665	0	150.980	173.35	-20.304	1
150.240	225.55	-21.989	1	151.000	172.99	-17.612	1
150.260	225.08	-19.294	1	151.020	170.77	-19.505	0
150.280	223.67	-19.593	0	151.040	168.37	-21.334	0
150.300	221.08	-22.830	0	151.060	168.02	-18.641	1
150.320	220.62	-20.135	1	151.080	165.72	-20.399	0
150.340	220.16	-17.443	1	151.100	164.33	-19.907	1
150.360	217.80	-20.592	0	151.120	163.04	-19.396	0
150.380	216.28	-20.792	1	151.140	161.71	-18.870	1
150.400	215.82	-18.100	1	151.160	161.37	-16.180	1
150.420	213.41	-21.132	0	151.180	161.03	-13.494	1
150.440	211.88	-21.273	1	151.200	160.70	-10.811	1
150.460	211.43	-18.580	1	151.220	160.36	-8.131	1
150.480	208.99	-21.494	0	151.240	159.66	-7.587	0
150.500	208.55	-18.800	1	151.260	158.93	-7.036	1
150.520	207.21	-18.880	0	151.280	158.60	-4.361	1
150.540	204.75	-21.680	0	151.300	158.27	-1.691	1

Okno 3: Stan ustalony po skoku pod obciążeniem (160 V, 538 W, t □ [170.0, 171.5] ms, krok 20 µs)

t [ms]	v_C [V]	i_L [A]	s	t [ms]	v_C [V]	i_L [A]	s
170.000	160.24	3.618	0	170.760	160.23	4.352	0
170.020	159.90	6.278	1	170.780	160.09	4.876	1
170.040	159.93	6.804	0	170.800	159.76	7.535	1
170.060	160.19	5.194	0	170.820	160.09	5.927	0
170.080	160.30	3.582	0	170.840	160.27	4.315	0
170.100	159.96	6.242	1	170.860	160.13	4.839	1
170.120	159.98	6.767	0	170.880	159.80	7.498	1
170.140	160.25	5.156	0	170.900	160.13	5.890	0
170.160	160.35	3.542	0	170.920	160.31	4.277	0
170.180	160.01	6.202	1	170.940	160.16	4.800	1
170.200	160.22	4.592	0	170.960	159.83	7.459	1
170.220	160.09	5.116	1	170.980	160.16	5.850	0
170.240	160.06	5.642	0	171.000	160.33	4.236	0
170.260	160.21	4.031	0	171.020	160.19	4.760	1
170.280	159.87	6.691	1	171.040	159.85	7.418	1
170.300	160.13	5.083	0	171.060	160.18	5.809	0
170.320	160.22	3.472	0	171.080	160.34	4.195	0
170.340	159.88	6.132	1	171.100	160.20	4.718	1

t [ms]	v_C [V]	i_L [A]	s	t [ms]	v_C [V]	i_L [A]	s
170.360	159.90	6.659	0	171.120	159.86	7.377	1
170.380	160.15	5.050	0	171.140	160.18	5.767	0
170.400	160.24	3.439	0	171.160	160.35	4.153	0
170.420	159.91	6.099	1	171.180	160.20	4.677	1
170.440	159.92	6.626	0	171.200	159.86	7.336	1
170.460	160.17	5.016	0	171.220	160.18	5.726	0
170.480	160.26	3.405	0	171.240	160.34	4.112	0
170.500	159.92	6.065	1	171.260	160.19	4.636	1
170.520	159.94	6.591	0	171.280	159.85	7.295	1
170.540	160.18	4.982	0	171.300	160.17	5.685	0
170.560	160.26	3.370	0	171.320	160.32	4.072	0
170.580	159.93	6.030	1	171.340	160.17	4.596	1
170.600	159.94	6.557	0	171.360	159.83	7.255	1
170.620	160.18	4.947	0	171.380	160.14	5.646	0
170.640	160.26	3.335	0	171.400	160.29	4.034	0
170.660	159.93	5.996	1	171.420	160.14	4.558	1
170.680	159.94	6.522	0	171.440	159.80	7.217	1
170.700	160.17	4.913	0	171.460	160.11	5.609	0
170.720	160.06	5.437	1	171.480	160.26	3.998	0
170.740	160.05	5.963	0	171.500	160.10	4.522	1